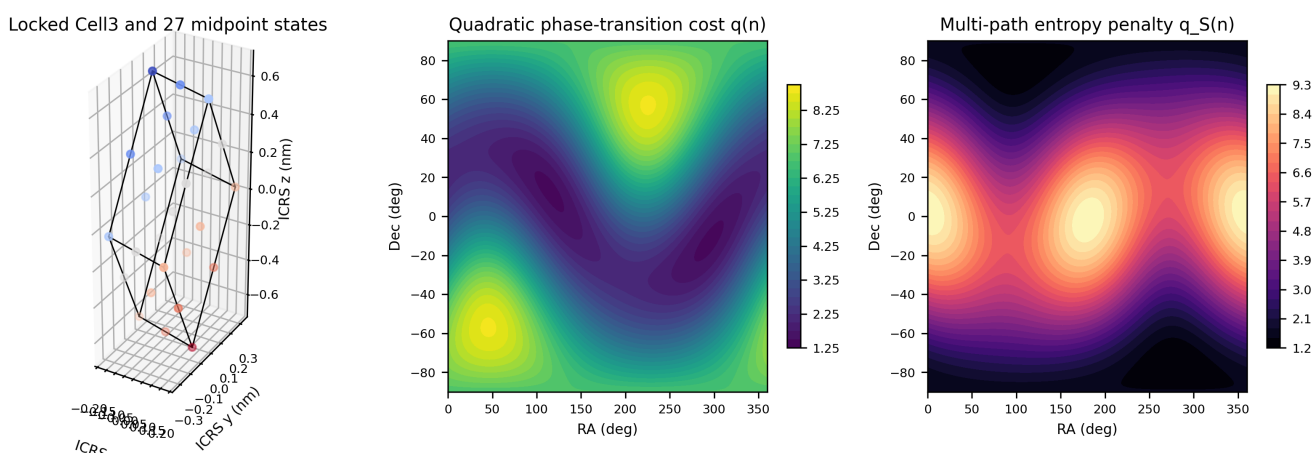


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K-FIELD ENTROPY VERSUS QUADRATIC PHASE-TRANSITION COST SURFACE

A foundational reference on exact scalar endpoint addressing, native multi-path entropy, quadratic directional cost, cost-entropy competition, precessional residence, and curvature-organized occupancy

K-Field endpoint geometry, directional cost, and path entropy



Principal Investigator

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Krytronx

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Classification and use contract

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This reference presents the K-Field endpoint-address, path-entropy, and phase-cost structures as one internally self-sufficient mathematical system. Exact source values, exact enumerations, derived matrices, continuum limits, and measured curvature statistics are kept distinct by their declared definitions and units.

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Abstract

The K-Field provides one explicit signed scalar address for every admissible location relative to a selected start point. The address is deterministic and lossless: it decodes to one integer lattice vector and therefore to one physical Cell3 displacement, one laminar grade, one polarity, and one recursive address. At the same time, an endpoint generally has many equal-weight admissible histories. The endpoint is certain while the route is uncertain. This route multiplicity produces a native combinatorial entropy that does not arise from measurement error or incomplete state description.

The same corpus defines a quadratic phase-transition cost surface on projective orientation space. For a unit ICRS direction $\mathbf{n}$, $q(\mathbf{n}) = \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}$. The soft, saddle, and hard eigenaxes give the stationary orientations and the anisotropic cost spread. The cost surface and the path entropy are separate native properties. Cost assigns directional resistance; entropy counts how many admissible histories feed an explicitly addressed endpoint.

Combining these structures gives the exact endpoint weight $\Omega_N(\mathbf{k}) \exp[-\beta_K q(\mathbf{k})]$ and the effective action $\beta_K q(\mathbf{k}) - \ln \Omega_N(\mathbf{k})$. In the large-event limit, equal-weight path multiplicity generates a second quadratic orientation surface $q_S(\mathbf{n}) = \mathbf{n}^T \mathbf{G}_S \mathbf{n}$, where $\mathbf{G}_S = \mathbf{B}^T(-\mathbf{T})\mathbf{B}(-1)$. The phase-cost and entropy surfaces are not coaxial. Their competition rotates the preferred effective orientation and demonstrates why observed organization follows minimum effective cost rather than minimum phase cost alone. The exact 10-event experiment confirms that the unbiased path entropy is substantial but does not itself reproduce the independent phase-cost surface. The two structures must be combined rather than conflated.

Foundational statement

The K-Field state space is simultaneously explicit and statistical. Every endpoint has one exact scalar address, while the set of equal-weight paths reaching that endpoint may be highly degenerate. Entropy is therefore an intrinsic property of K-Field propagation. The quadratic phase-transition cost acts on this entropy-bearing propagation ensemble and organizes directional occupancy together with residence time and curvature focusing.

1. Coordinate, dimensional, and state-space contract

1.1 ICRS physical frame

All physical vectors are expressed as Cartesian columns in the International Celestial Reference System. The x axis points toward right ascension 0 degrees and declination 0 degrees. The y axis points toward right ascension 90 degrees and declination 0 degrees. The z axis points toward the north celestial pole. This fixed frame prevents hidden rotations or relabeling when the direct Cell3 geometry, the phase-cost surface, and the entropy surface are compared.

1.2 Distinct spaces and units

Object	Definition	Units and role
Address $\mathbf{k}$	integer triple in $\mathbb{Z}^3$	exact relational location relative to a start point
Physical displacement $\mathbf{r}$	$\mathbf{r} = \mathbf{B}\mathbf{k}$	nm; direct Cell3 geometry
Unit direction $\mathbf{n}$	$\mathbf{n} = \mathbf{r} / \ \mathbf{r}\ $	dimensionless projective orientation
Serialized address ρ	$\rho(\mathbf{k}) = \text{sign}(\mathbf{k}) \sqrt{\mathbf{k}^T \mathbf{G} \mathbf{k}}$	one signed scalar that uniquely identifies $\mathbf{k}$
Phase cost q	$q(\mathbf{n}) = \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}$	dimensionless directional transition cost
Path multiplicity Ω_N	number of admissible N-event histories reaching $\mathbf{k}$	integer degeneracy
Path entropy S_N	$S_N = \mathbf{k}_B \ln \Omega_N$	entropy associated with route uncertainty
Entropy penalty q_S	$q_S(\mathbf{n}) = \mathbf{n}^T \mathbf{G}_S \mathbf{n}$	nm^{-2} ; directional suppression of path multiplicity at fixed radius

The address, the physical displacement, the orientation, the cost, and the entropy are related but are not interchangeable. The signed scalar determines the endpoint. Path multiplicity describes the number of admissible histories compatible with that endpoint. Phase cost evaluates the endpoint direction. The unit declarations prevent a scalar cost from being added directly to an entropy term until a conversion coefficient is supplied.

2. Locked Cell3 geometry, midpoint nodes, and boundary planes

2.1 Ordered basis and direct metric

The physical Cell3 is generated by the ordered basis $B=[a \ b \ c]$. The column order is structural because it fixes the three primitive transition families and their direct-reciprocal correspondence. The direct metric is $G=B^T B$, and the physical cell volume is $|\det B|$.

```

B (nm) =
[ 0.327501242069651  0.035330775844051 -0.010129317626665]
[-0.054897007886313  0.207319850441498 -0.389787702363804]
[-0.037487502896305 -0.547401085391957 -0.689734408362922]

G = B^T B (nm^2) =
[ 0.111676057905435  0.020710333281606  0.043937235097789]
[ 0.020710333281606  0.343877732397120  0.296392758961523]
[ 0.043937235097789  0.296392758961523  0.627770610069385]

```

Quantity	Value
a	0.334179679073151 nm
b	0.586410890414836 nm
c	0.792319765037693 nm
A_ab	0.194869165469900 nm^2
A_ac	0.261106235799531 nm^2
A_bc	0.357809539128562 nm^2
Volume	0.117914891916699 nm^3

Dimensionally correct Cell3: 27 nodes and six primitive boundary planes

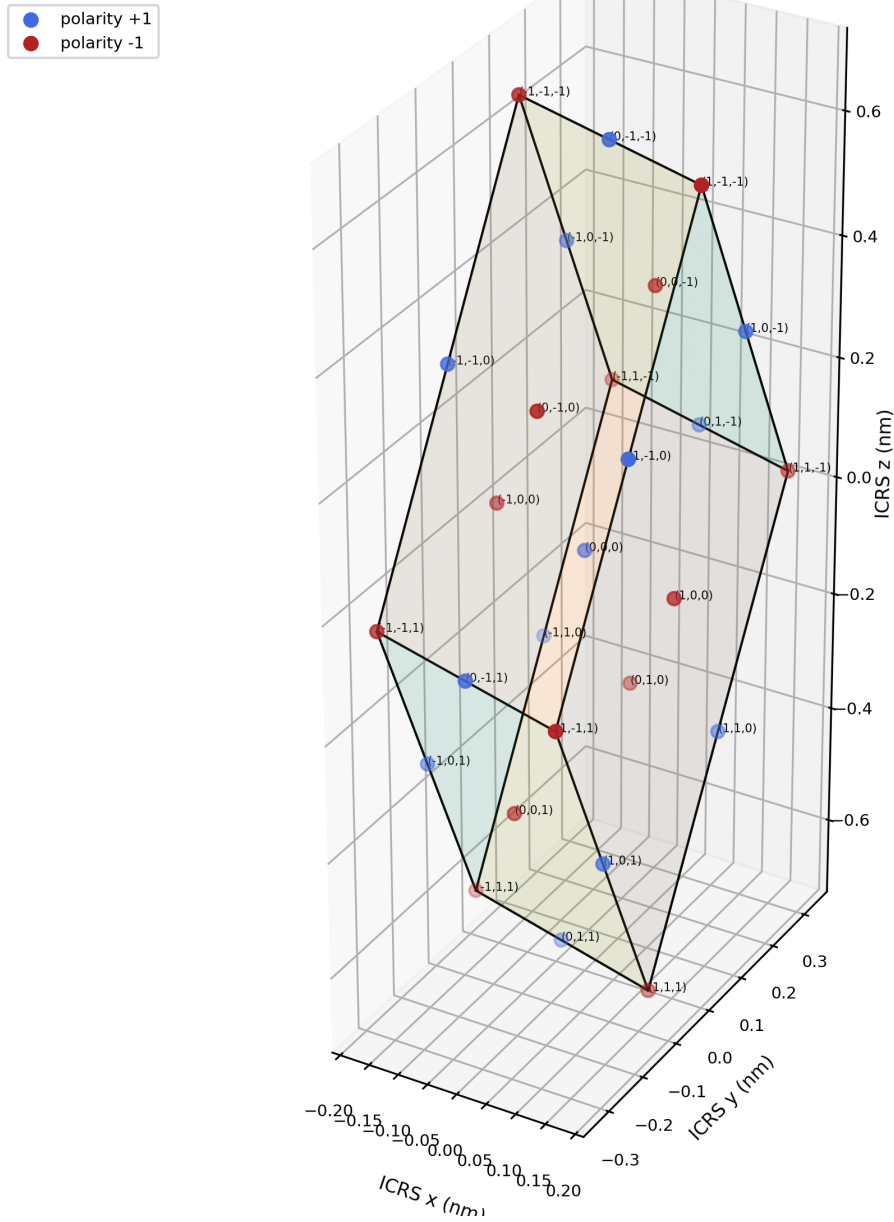


Figure 1. Dimensionally correct ICRS Cell3 with all 27 midpoint nodes, exact address labels, checkerboard polarity, and all six signed primitive boundary planes.

2.2 The complete 27-node midpoint state set

The midpoint address set is $L=\{-1,0,1\}^3$. Each node has physical position $p(L)=BL/2$. The laminar grade is $\lambda=i+j+k$ and checkerboard polarity is $\chi=(-1)^\lambda$. This creates seven sheets with populations 1, 3, 6, 7, 6, 3, and 1. The node kind is determined by the number of nonzero address components: three gives a corner, two an edge center, one a face center, and zero the body center.

Seven-sheet Cell3 laminar stack: populations 1, 3, 6, 7, 6, 3, 1

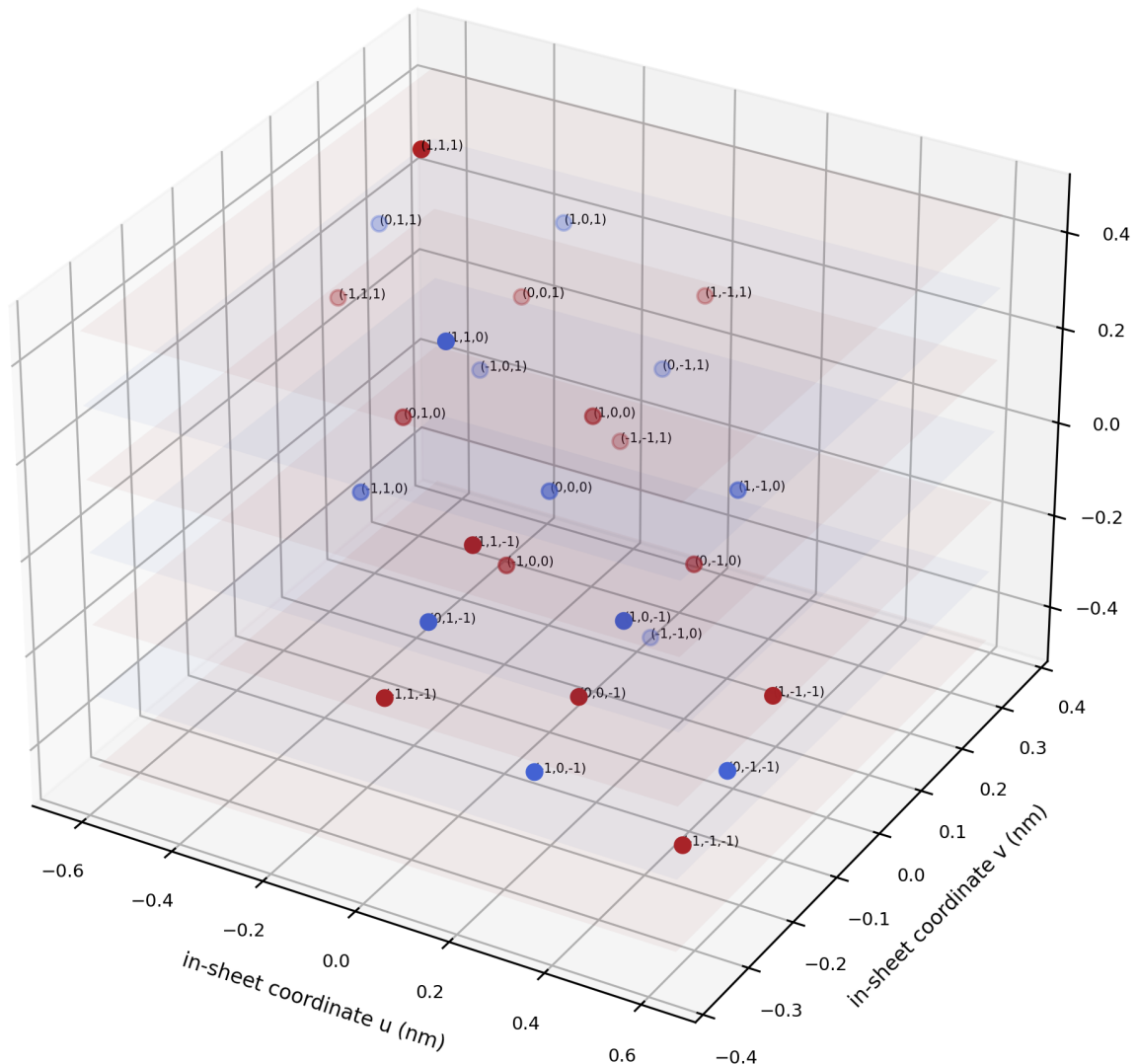


Figure 2. The exact seven-sheet decomposition of the 27 midpoint states. Every sheet has one polarity; each address is labeled.

2.3 Six primitive boundary planes

Each primitive face family has two signed boundary planes. Fixing address coefficient $a=+1$ or $a=-1$ selects the two bc planes; fixing b selects the ac planes; fixing c selects the ab planes. Every plane contains exactly nine midpoint nodes. The face areas and opposite spacings are generated from the same ordered basis and preserve the full oblique geometry.

All six primitive boundary planes and all nine midpoint nodes per plane

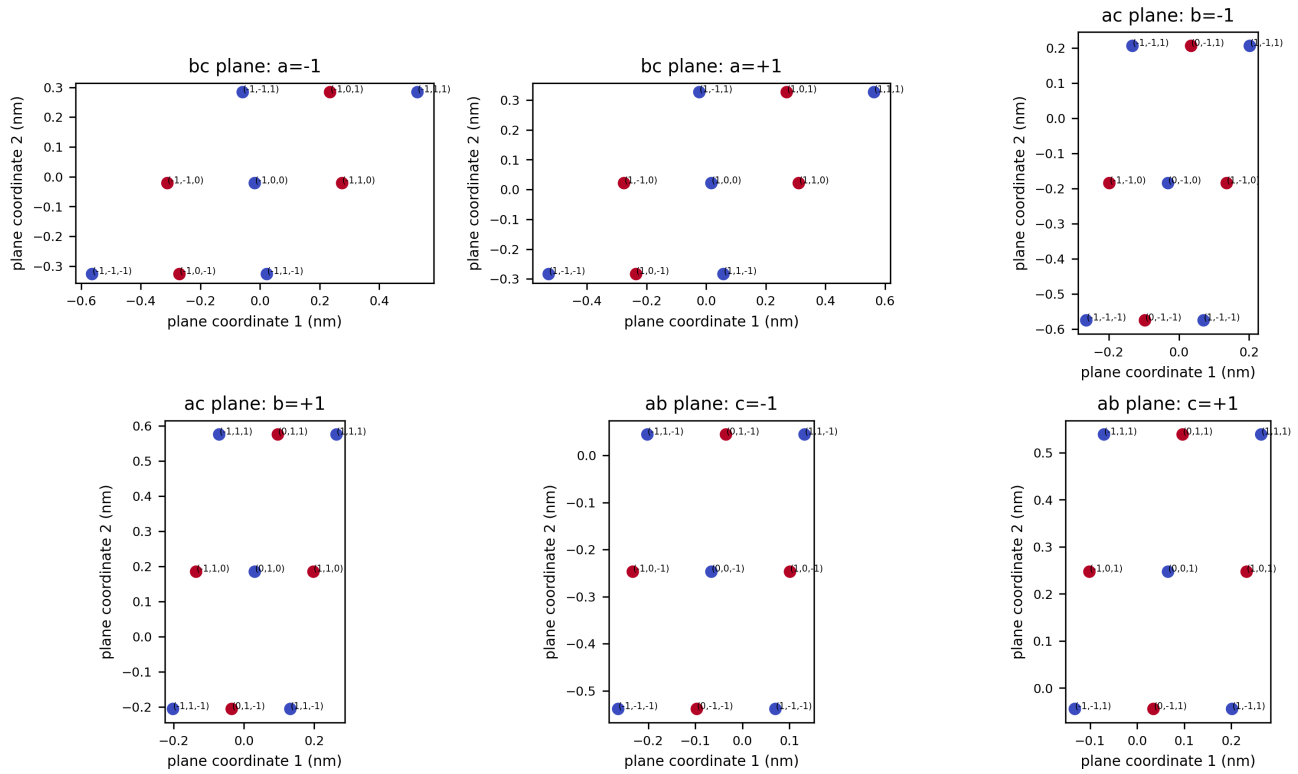


Figure 3. Six-panel atlas of the signed primitive boundary planes. Each panel preserves true in-plane metric and labels all nine member nodes.

3. Lossless one-dimensional serialized spectral addressing

3.1 Central address theorem

Let the relational lattice be $\Lambda = B \mathbb{Z}^3$ and define the direct quadratic shell $Q(k) = k^T G k$. The signed scalar address is $\rho(k) = \sigma(k) \sqrt{Q(k)}$, where the sign function supplies directed antipodal ordering. Under the corpus injectivity condition $Q(k) = Q(l)$ only when $l = +k$ or $l = -k$, the signed coordinate is a bijection: $\rho(k) = \rho(l)$ implies $k = l$.

$$\begin{aligned} \Lambda &= B \mathbb{Z}^3 \\ Q(k) &= k^T G k \\ \rho(k) &= \sigma(k) \sqrt{Q(k)} \\ \rho(k) &= \rho(l) \Rightarrow k = l \end{aligned}$$

One scalar therefore carries the complete directed endpoint address. Decoding ρ recovers k , and k recovers the physical displacement Bk , laminar grade, polarity, projective channel, and balanced-ternary recursive digits. The scalar coordinate serializes the hidden rank-three relational geometry rather than collapsing it.

3.2 Native scalar arithmetic

Addition and subtraction are performed by decoding, operating in the integer module, and re-encoding. This preserves the relational geometry even though the explicit coordinate is one-dimensional.

$$\begin{aligned} x (+) y &= \rho[\rho^{(-1)}(x) + \rho^{(-1)}(y)] \\ x (-) y &= \rho[\rho^{(-1)}(x) - \rho^{(-1)}(y)] \\ _G &= [x^2 + y^2 - (x (-) y)^2] / 2 \end{aligned}$$

3.3 Endpoint certainty does not remove path uncertainty

A unique endpoint address does not imply a unique history. The documented address (4,-2,2) has one scalar endpoint and 420 shortest paths. The paths consist of four +a events, two -b events, and two +c events in every distinct ordering. The multiplicity is $8!/(4!2!2!)=420$.

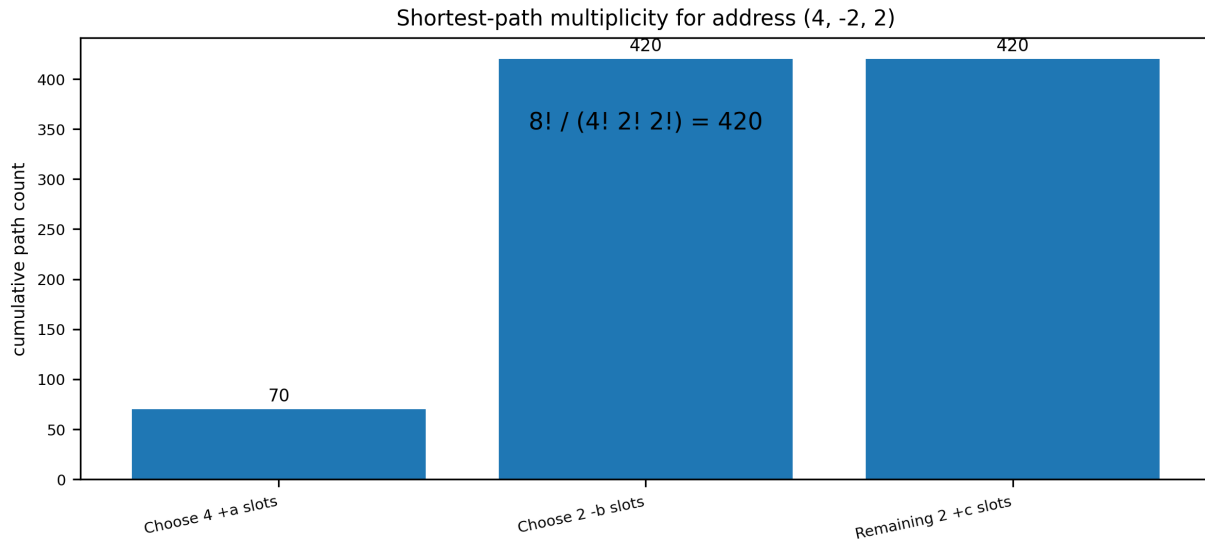


Figure 4. The documented serialized endpoint (4,-2,2) has 420 equal-length shortest paths, obtained by exact multinomial ordering.

4. Native multi-path propagation and entropy

4.1 Equal-weight primitive transition alphabet

The exact 10-event propagation experiment begins at Cell3 corner address (1,1,1) and uses the six directed primitive moves +a, -a, +b, -b, +c, and -c. Each event has probability 1/6. The tiling is unbounded. Exact dynamic programming enumerates all $6^{10}=60,466,176$ ordered histories without Monte Carlo sampling.

4.2 Endpoint fibers

For N events, $\Gamma_N(k)$ is the set of admissible ordered histories ending at address k, and $\Omega_N(k)=|\Gamma_N(k)|$ is the fiber multiplicity. Equal-weight histories give conditional path probability $1/\Omega_N(k)$ after the endpoint is known.

```

Gamma_N(k) = {ordered admissible N-event paths ending at k}
Omega_N(k) = |Gamma_N(k)|
P(path | k, N) = 1 / Omega_N(k)
S_N(k) = -k_B ln Omega_N(k)

```

This entropy is not an uncertainty in the endpoint. It is the exact logarithmic size of the path fiber above an explicit endpoint. The K-Field is therefore deterministic in address and non-deterministic in propagation.

4.3 Exact entropy chain rule

Entropy quantity	Exact value
History entropy	17.917594692280549 nats
Endpoint entropy	5.364728188272016 nats
Mean conditional fiber entropy	12.552866504008534 nats
Chain-rule residual	-1.776e-15 nats
Reachable endpoint count	891
Effective endpoint count exp(H)	213.733131612919
Effective fiber multiplicity exp(H)	282905.020591319306

The fraction of total propagation entropy remaining after the endpoint is known is 70.058882 percent. Path uncertainty is therefore the dominant part of the 10-event information budget.

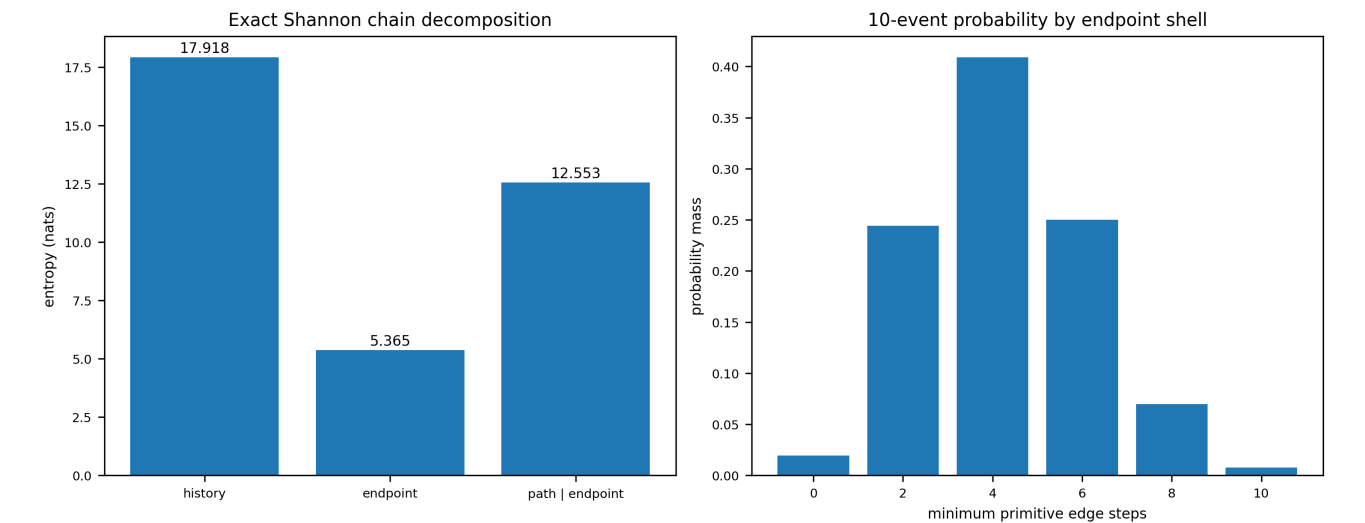


Figure 5. The 10-event history entropy separates exactly into endpoint entropy and mean conditional path-fiber entropy. The right panel shows probability by minimum-step shell.

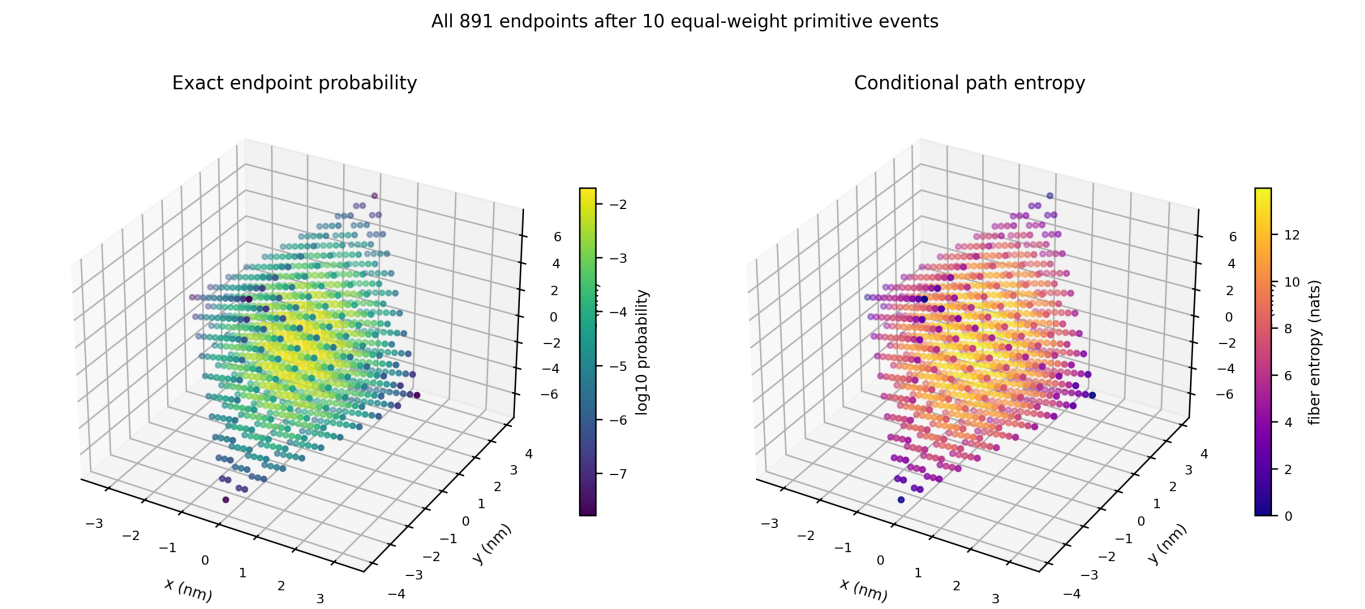


Figure 6. Exact dynamic-programming endpoint distribution for all 60,466,176 ordered histories. Position is explicit; color records probability or path-fiber entropy.

5. Quadratic phase-transition cost surface

5.1 Directional cost field

The K-Field directional cost is the positive-definite quadratic form $q(n)=n^T G_{\text{phase}} n$ restricted to unit ICRS directions. Because n and $-n$ have the same cost, the natural orientation space is projective. The surface is a scalar field on orientation space, not a physical ellipsoid in position space.

```
G_phase =
[ 2.976410637800  1.243150779500 -2.080360035400]
[ 1.243150779500  2.592228663500 -2.317883993700]
[-2.080360035400 -2.317883993700  6.773570913300]

q(n) = n^T G_phase n,  ||n|| = 1
```

Axis	q eigenvalue	RA (deg)	Dec (deg)	q_S at phase axis (nm ⁻²)
soft	1.429328753487	114.813083651	12.969895704	5.738931542435
saddle	2.112545026447	197.222212119	-29.836687522	7.103047066506
hard	8.800336434665	225.519007199	56.919969623	4.000616506795

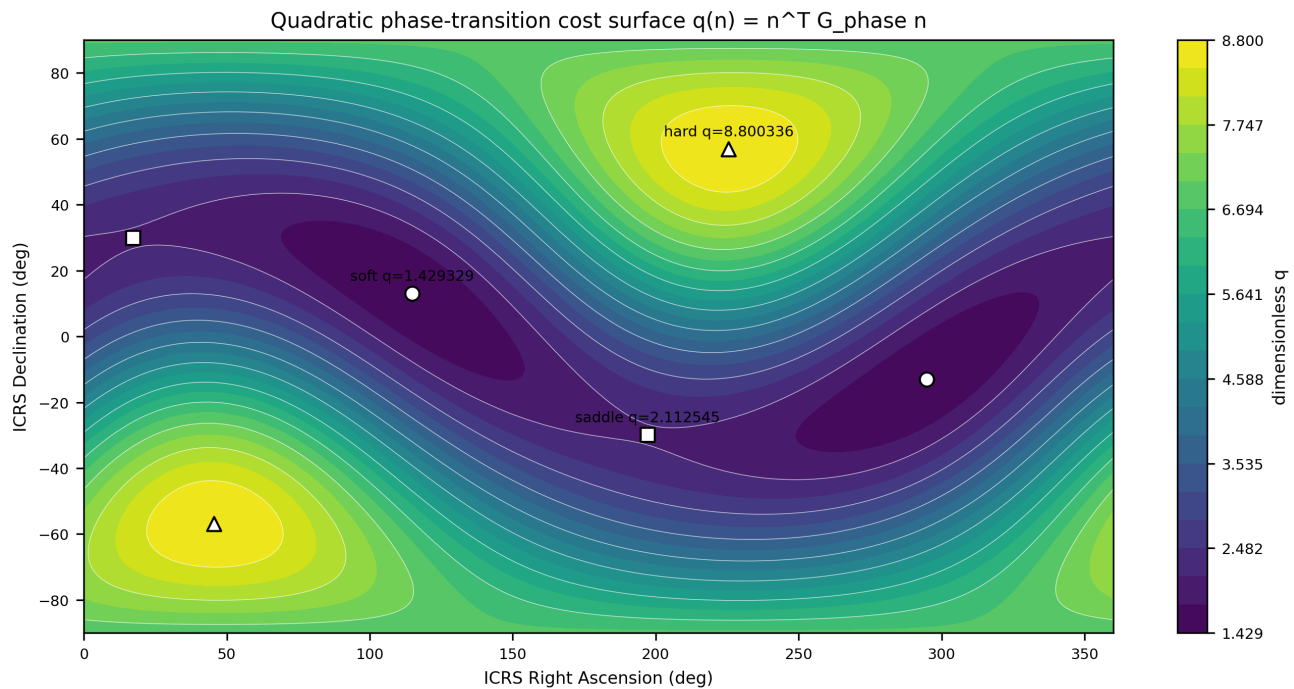


Figure 7. Exact ICRS quadratic phase-transition cost surface with soft, saddle, and hard stationary directions and their antipodes.

5.2 Address-energy representation

The phase matrix is generated from the local address-energy coupling matrix W through the exact identity $G_{\text{phase}}=4B^T(-T)WB^T(-1)$. This relation shows that the directional cost in ICRS space is the physical image of a quadratic address-space form.

```
G_phase = 4 B^T(-T) W B^T(-1)
q(k) = [k^T B^T G_phase B k] / [k^T G k]
      = 4 [k^T W k] / [k^T G k]
```

Because k is decoded from ρ , the phase cost is an explicit scalar-address observable: $q(x)=q[\rho^T(-1)(x)]$. No additional orientation state is needed.

6. Cost and entropy are complementary K-Field properties

6.1 The natural endpoint weight

Equal-weight path multiplicity and directional phase cost combine multiplicatively. For an endpoint k reached in N events, the unnormalized state weight is $\Omega_N(k) \exp[-\beta_K q(k)]$. The coefficient β_K converts the dimensionless phase cost into logarithmic statistical weight and is common to the complete ensemble.

$$\begin{aligned} w_N(k) &= \Omega_N(k) \exp[-\beta_K q(k)] \\ A_N(k) &= \beta_K q(k) - \ln \Omega_N(k) \end{aligned}$$

The effective action A_N is lower when the phase cost is lower or when the number of admissible paths is larger. A high-cost endpoint can dominate if it has enough path multiplicity. A low-cost endpoint can remain weakly occupied if only a small path fiber reaches it.

6.2 Exact phase-selection boundary

Two endpoint families i and j have equal effective weight when their phase-cost difference equals their entropy difference in logarithmic units.

$$\begin{aligned} A_i &= A_j \\ \beta_K (q_i - q_j) &= \ln[\Omega_i / \Omega_j] \\ &= (S_i - S_j) / k_B \end{aligned}$$

This equality is the exact cost-entropy transition condition. It defines where a preferred endpoint family changes as phase weighting, propagation depth, or path multiplicity changes.

7. Exact 10-event test of phase cost versus path entropy

7.1 Direct endpoint comparison

The phase cost was evaluated for every nonzero endpoint in the exact 10-event destination registry. The endpoint unit direction is $\mathbf{n} = \mathbf{B}_k / \|\mathbf{B}_k\|$, and $q = \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}$. This produces 890 nonzero endpoint comparisons with no sampling.

Statistic	Value
Nonzero endpoints	890
Equal-endpoint corr(q , fiber entropy)	-0.034449777988
Probability-weighted corr(q , fiber entropy)	0.001368924044
Probability-weighted mean q	4.696481203521
Probability-weighted mean fiber entropy	12.524749223618 nats
Unweighted q regression coefficient after distance and step count	0.018379535576
Weighted q regression coefficient after distance and step count	0.004413309583

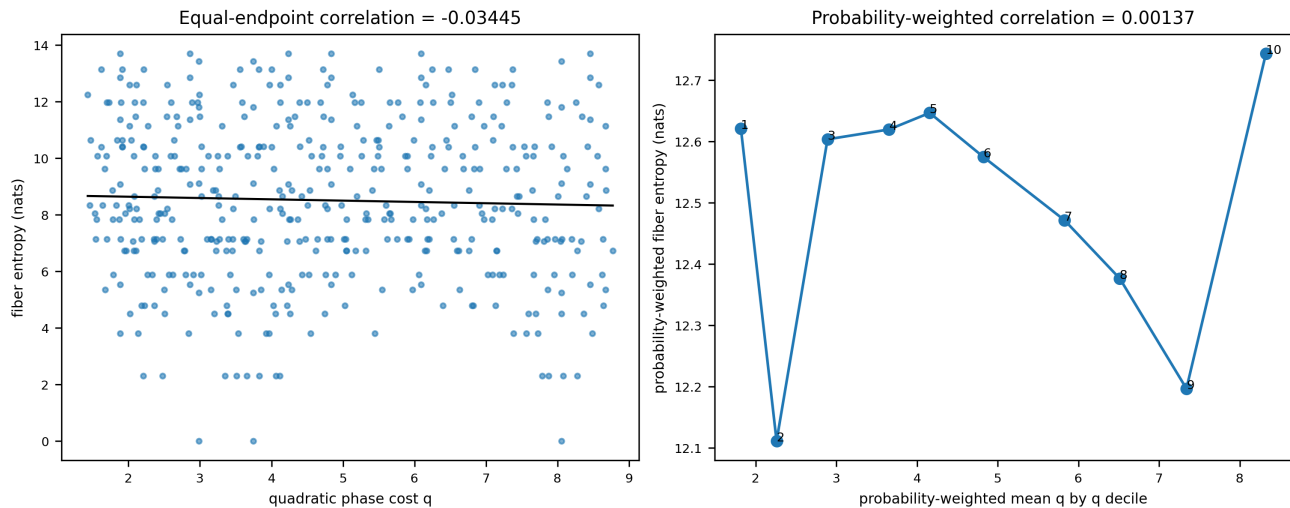


Figure 8. Exact 10-event comparison. The equal-weight propagation baseline carries substantial path entropy but no material correlation with the independent phase-cost surface.

7.2 Interpretation of the near-zero correlation

The exact unbiased propagation experiment assigns probability 1/6 to each primitive transition and does not use G_{phase} when selecting events. Its path entropy therefore measures combinatorial route degeneracy only. The near-zero correlation is not a failure of either structure. It demonstrates that native path entropy and the quadratic phase-cost surface are independent inputs to the complete K-Field population law.

If the cost surface had already been implicit in the equal-weight path count, a strong correlation would appear without any cost weighting. It does not. The phase surface must act on the entropy-bearing endpoint ensemble rather than being identified with the entropy itself.

8. The entropy-generated quadratic orientation surface

8.1 Large-event equal-weight limit

For the six equal-probability primitive moves, the endpoint distribution approaches a three-dimensional Gaussian in address space. The leading exponent is $-(3/2N)k^{\wedge}T$ k. The exact finite-N distribution retains integer and parity structure, while the Gaussian gives the continuous directional law.

$$P_N(k) = \Omega_N(k) / 6^{\wedge}N$$

$$P_N(k) \text{ proportional to } \exp[-(3/(2N)) k^{\wedge}T k]$$

8.2 Transformation into physical ICRS space

At fixed physical radius R , write $r=Rn$ and $k=B^{\wedge}(-1)r$. Then $k^{\wedge}T k=R^{\wedge}2 n^{\wedge}T B^{\wedge}(-T)B^{\wedge}(-1)n$. This produces a second quadratic surface on orientation space.

$$G_S = B^{\wedge}(-T) B^{\wedge}(-1)$$

$$q_S(n) = n^{\wedge}T G_S n$$

$$\ln \Omega_N(R, n) = \text{constant} - [3R^{\wedge}2/(2N)] q_S(n) + \text{higher-order terms}$$

$$G_S \text{ (nm}^{\wedge}-2) =$$

9.275643997185	0.095769243769	0.274708806720
0.095769243769	6.024094143790	-1.217784618059
0.274708806720	-1.217784618059	1.542856974761

Axis	q_S eigenvalue (nm ⁻²)	RA (deg)	Dec (deg)
soft	1.222872851451	98.306312947	75.582702912
saddle	6.333432713989	270.502388945	14.289263978

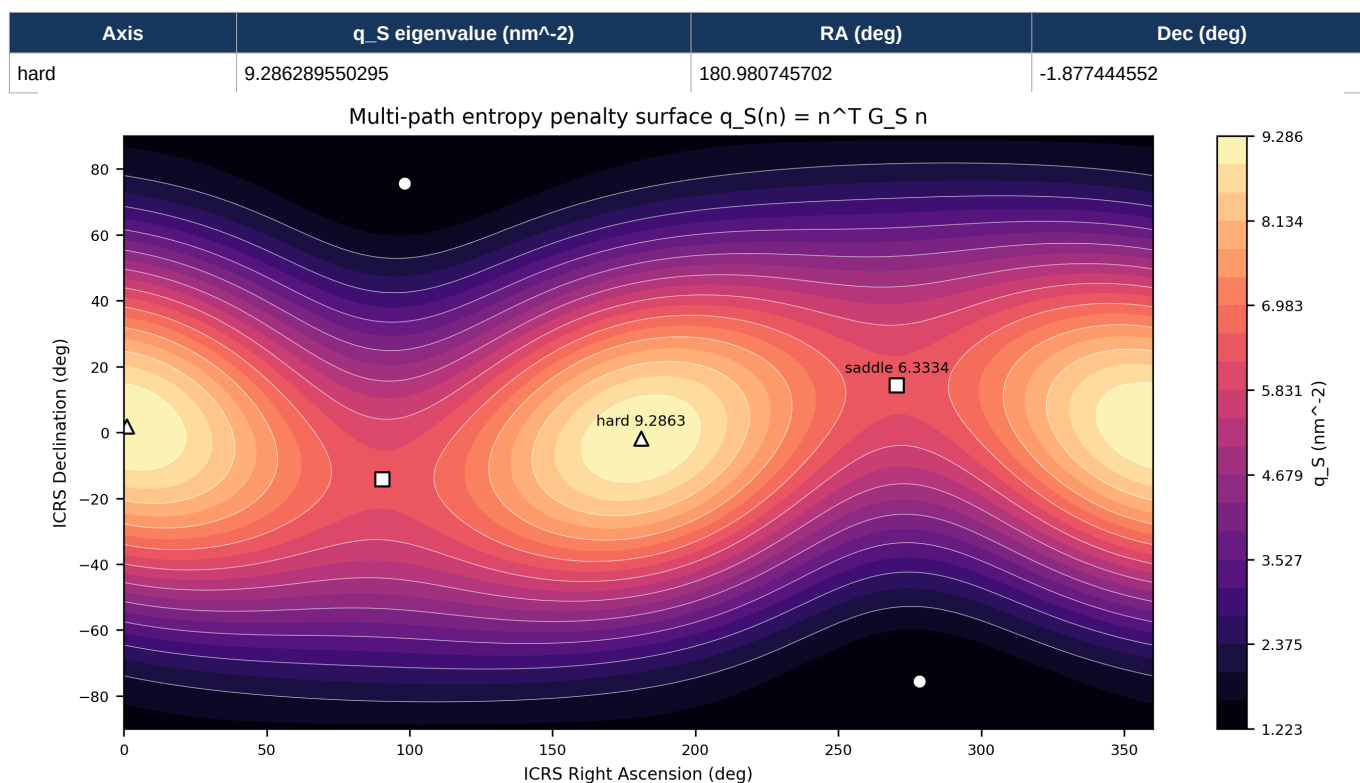


Figure 9. Derived physical-orientation penalty associated with equal-weight address-space diffusion. Lower q_S means greater path multiplicity at fixed physical radius.

The entropy surface is anisotropic in physical space even though the six moves are equiprobable in address space. The anisotropy arises because B maps equal address increments into unequal oblique physical vectors.

9. Misalignment and competition between the two quadratic surfaces

9.1 Non-coaxial eigensystems

G_phase and G_S do not share eigenvectors. The phase axes and entropy axes occupy different ICRS directions. Therefore the combined effective orientation cannot be reduced to rescaling one fixed ellipsoid. Changing the entropy-to-cost ratio rotates the effective eigenaxes.

Phase axis / Entropy axis	E-soft	E-saddle	E-hard
P-soft	63.256214 deg	36.369845 deg	67.282432 deg
P-saddle	58.980169 deg	83.163430 deg	31.931459 deg
P-hard	43.169375 deg	54.484404 deg	68.814817 deg

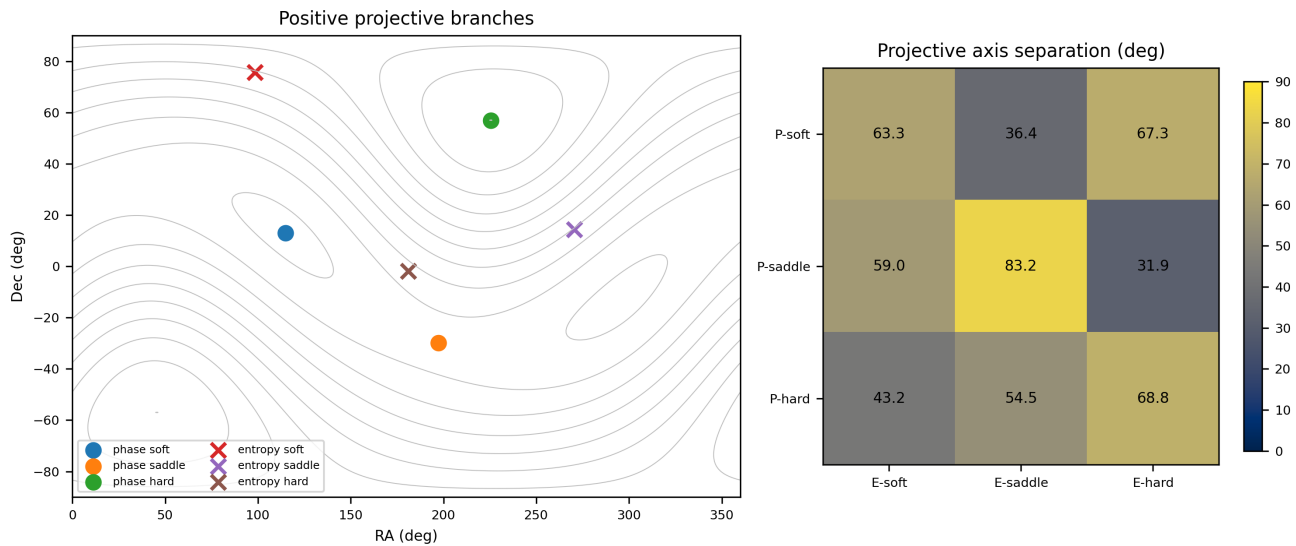


Figure 10. Comparison of the phase-cost and entropy-penalty eigensystems. The two quadratic surfaces are not coaxial.

9.2 Phase-axis comparison

Phase axis	q	q_S (nm ⁻²)	Cost role	Entropy role at fixed R
soft	1.429328753	5.738931542	minimum phase cost	intermediate entropy accessibility
saddle	2.112545026	7.103047067	intermediate phase cost	largest penalty among phase axes
hard	8.800336435	4.000616507	maximum phase cost	smallest penalty among phase axes

The phase-hard axis is strongly penalized by q but is comparatively favored by path entropy at fixed physical radius because its q_S value is lower than the q_S values at the phase-soft and phase-saddle axes. This establishes a direct competition: phase cost suppresses the hard direction while multi-path entropy partly restores it.

10. Effective directional population matrix

10.1 Combined large-event action

Substituting the asymptotic path multiplicity into the cost-entropy action gives one quadratic effective orientation matrix.

$$A_N(R, n) = n^T [\beta_K G_{\text{phase}} + 3R^2 G_S / (2N)] n + \text{constant}$$

$$G_{\text{eff}} = \beta_K G_{\text{phase}} + 3R^2 G_S / (2N)$$

Using $G_{\text{phase}} = 4B^{(-T)}WB^{(-1)}$ and $G_S = B^{(-T)}IB^{(-1)}$, the direct address-space form is

$$G_{\text{eff}} = B^{(-T)} [4\beta_K W + 3R^2 I / (2N)] B^{(-1)}$$

The matrix W supplies phase-transition anisotropy. The identity matrix supplies equal-weight address-space diffusion. $B^{(-1)}$ maps both structures into physical ICRS orientation space. The ratio $c = [3R^2 / (2N)] / \beta_K$ has units of nm² and controls the competition.

10.2 Crossover scales evaluated on the phase axes

Axis pair	Delta q	Delta q_S (nm ⁻²)	crossover c (nm ²)
soft vs saddle	0.683216272960	1.364115524071	none for positive c
soft vs hard	7.371007681178	-1.738315035640	4.240317508651
saddle vs hard	6.687791408218	-3.102430559711	2.155661917165

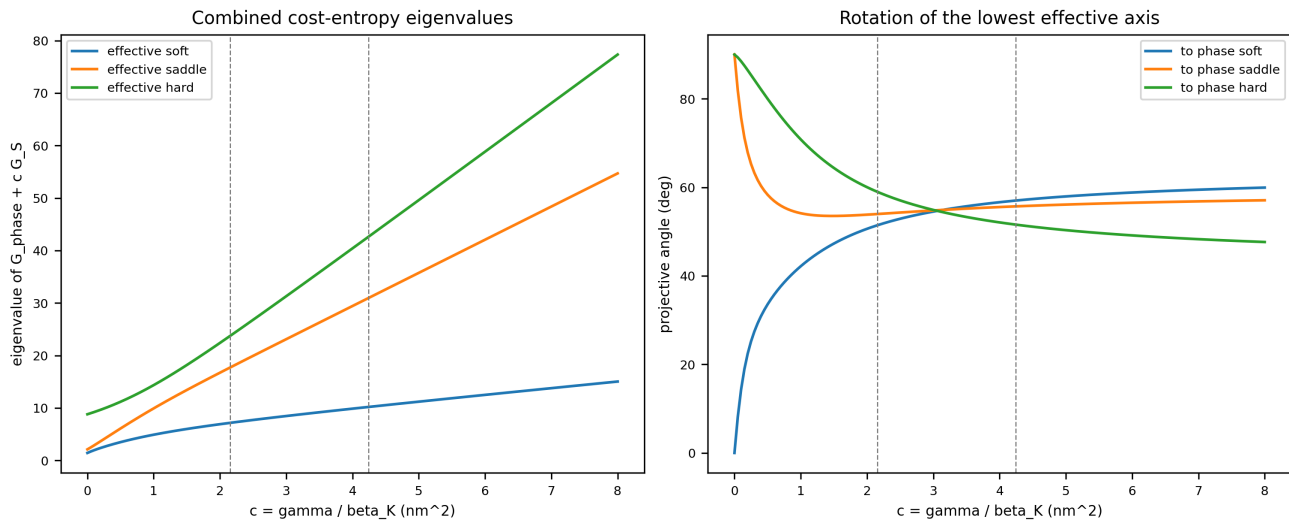


Figure 11. Parameter-free matrix sweep in $c = \gamma / \beta_K$ using the locked G_{phase} and G_S matrices. It shows how the preferred effective direction rotates as entropy weighting increases.

Because the matrices do not commute, the full effective eigenvectors rotate continuously rather than simply exchanging the original phase axes. The sweep is generated directly from the two locked matrices and requires no fitted endpoint coefficients.

11. Phase isolines, precession, and residence time

11.1 Cost-conserving orientation flow

For fixed angular-momentum magnitude L , the K-Field torque associated with the quadratic cost is tangent to the unit sphere. The orientation equation is $\dot{n} = -(2/L) n \times G_{\text{phase}} n$. The cross product preserves $\|n\|$ and gives $dq/dt = 0$. An undamped trajectory remains on one phase-cost isoline.

$$\begin{aligned} \tau_K &= -2 n \times G_{\text{phase}} n \\ \dot{n} &= -(2/L) n \times G_{\text{phase}} n \\ dq/dt &= 0 \\ \| \dot{n} \| &= (2/L) \| n \times G_{\text{phase}} n \| \end{aligned}$$

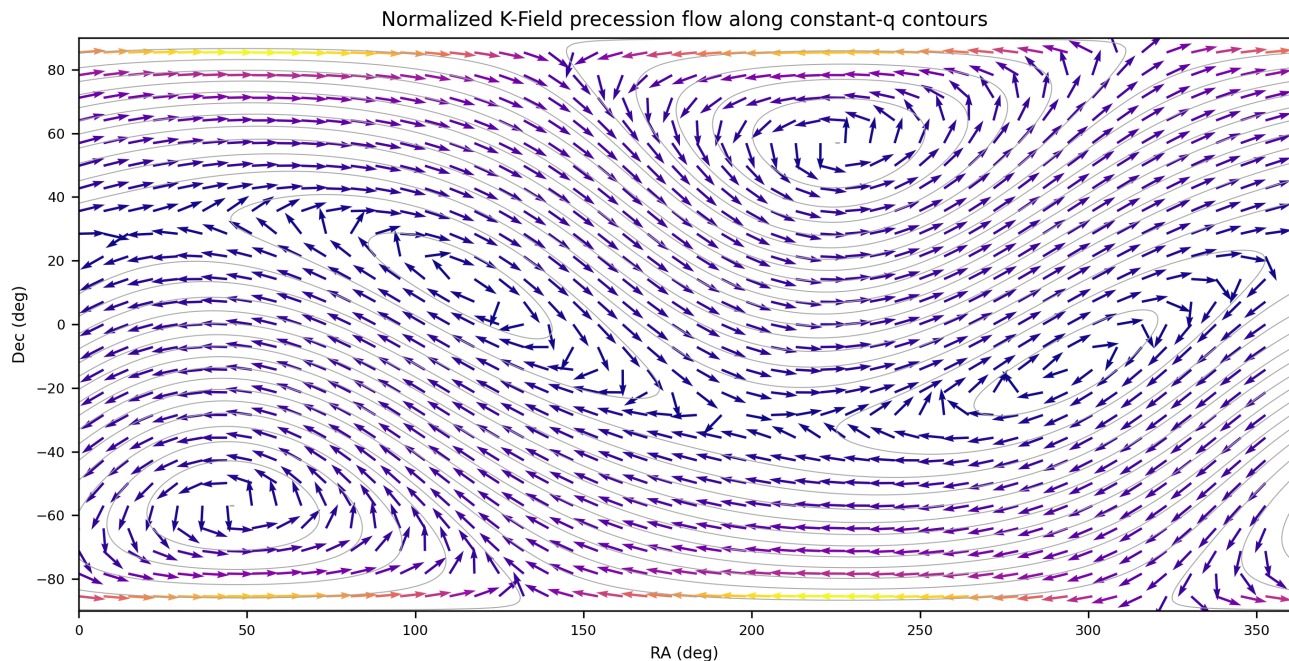


Figure 12. Orientation-space precession field generated from the exact G_{phase} matrix. The flow is tangent to the cost isolines and vanishes at the eigenaxes.

11.2 Dwell-time weighting

A mixed ensemble sampled at one time accumulates in slow sections of an orbit. The first-order residence density along an arc is proportional to $1/||\mathbf{n}_{\text{dot}}||$. This weight is distinct from path multiplicity: Ω_N counts discrete propagation histories reaching an endpoint, while dwell time measures how long orientation flow remains near a direction.

$$\begin{aligned} \text{snapshot density along one mixed orbit} &\propto dt/ds \\ &= 1 / ||\mathbf{n}_{\text{dot}}|| \end{aligned}$$

12. Curvature organization beyond scalar cost and speed

Equal phase cost does not imply equal contour spacing, turning rate, tangent curvature, or focusing of neighboring trajectories. The local derivative hierarchy therefore evaluates slope magnitude and the two signed curvature coordinates of the slope-magnitude surface. The corpus occupancy analysis shows that curvature carries independent organization after conditioning jointly on phase cost and slope magnitude.

Test	Statistic	Degrees of freedom	p value
Signed RA curvature deciles	G=22.9474	-	0.00632
Signed Dec curvature deciles	G=34.8533	-	6.324e-05
Full cubic vs additive cubic	Delta G=79.5962	3	3.747e-17
Curvature added after slope magnitude	Delta G=65.0276721	10	4.00419e-10
Curvature added after q and slope	Delta G=32.9748532	8	6.22341e-05
Covariant east-curvature deciles	G=70.8216	-	1.0506e-11

The independent curvature contribution means that occupancy cannot be reconstructed from a Boltzmann-like phase-cost factor alone. It requires the path-fiber measure, the cost weighting, the precessional residence factor, and local focusing or defocusing.

13. Unified K-Field population measure

13.1 Three native weighting layers

The complete population measure contains three structurally different factors. Path multiplicity counts how many admissible histories arrive. Phase selection weights the directional transition cost. Residence and curvature weight how orientation flow occupies the local phase portrait.

$$\begin{aligned} P_N(n) &\propto \Omega_N(n) \\ &\times \exp[-\beta_K q(n)] \\ &\times C(n) / ||\mathbf{n} \times G_{\text{phase}} \mathbf{n}|| \end{aligned}$$

$C(n)$ denotes the measured local curvature-focusing contribution. It is not a replacement for the other factors. Each factor answers a different question: how many paths arrive, how costly the direction is, and how long or how strongly the flow occupies the direction.

13.2 Endpoint certainty plus propagation entropy

The unified measure does not weaken endpoint determinism. The scalar address still identifies one exact location. Statistical behavior enters through the multiplicity and transport of admissible histories over the explicit address space.

14. Central findings

Finding	Exact or derived result
Explicit endpoint address	$\rho(k)$ uniquely identifies k and therefore B_k , λ , polarity, and recursive digits.
Native path entropy	$S_N(k) = k_B \ln \Omega_N(k)$ for equal-weight admissible histories.
10-event history count	60,466,176 exact ordered histories and 891 reachable endpoints.
Entropy decomposition	$17.9175946923 = 5.3647281883 + 12.5528665040$ nats.
Path fraction of information	70.0588818957 percent remains after the endpoint is known.
Phase cost	$q(n) = n^T G_{\text{phase}} n$ with eigenvalues 1.4293287535, 2.1125450264, and 8.8003364347.
Entropy surface	$q_S(n) = n^T B^*(-T)B^*(-1)n$ with eigenvalues 1.2228728514, 6.3334327140, and 9.2862895511 nm^{-2} .
Independence in unbiased propagation	$\text{corr}(q, H_{\text{fiber}}) = -0.03445$; probability-weighted $\text{corr} = 0.00137$.
Cost-entropy action	$A_N(k) = \beta_K q(k) - \ln \Omega_N(k)$.
Effective orientation matrix	$G_{\text{eff}} = \beta_K G_{\text{phase}} + \{3R^2/(2N)\} G_S$.
Non-coaxial competition	Changing the entropy-to-cost ratio rotates the preferred direction.
Complete occupancy	Path multiplicity $\times$ phase weight $\times$ dwell/focusing weight.

15. New pathways to discovery and direct tests

15.1 Event-count scaling

The entropy coefficient $3R^2/(2N)$ decreases as event count increases at fixed physical radius. The phase term does not carry this N dependence. The effective eigensystem should therefore evolve systematically with propagation depth. Measuring this rotation provides a direct separation of phase-cost and entropy contributions.

15.2 Radius scaling

At fixed event count, the entropy penalty grows as R^2 . Distant endpoints require increasingly specific address histories and therefore display stronger directional entropy anisotropy. A radius-resolved occupancy map can test the G_S surface independently of β_K .

15.3 Controlled transition bias

The exact 10-event baseline is unbiased. Introducing a measured directional phase transition law into the six primitive probabilities changes $\Omega_N(k)$ into a weighted path partition sum. Comparing unbiased and phase-weighted runs isolates the direct contribution of G_{phase} while preserving the explicit endpoint address.

15.4 Curvature-conditioned entropy test

Endpoints may be binned jointly by q , q_S , slope magnitude, and curvature state. If the unified measure is complete, adding q_S or exact finite- $N \ln \Omega$ should improve occupancy prediction after conditioning on q and derivatives. The improvement must use one common conversion law across all bins.

15.5 Recursive Lucas and Supergroup extension

The scalar address and balanced-ternary carry extend to Lucas Groups and Lucas Supergroups without changing the endpoint theorem. Multi-path entropy can therefore be computed at each recursive depth. The phase matrix remains locked while the path partition expands. This produces a direct program for identifying entropy-driven band populations and closure transitions across the complete hierarchy.

16. Reference formulas

Concept	Formula
Physical displacement	$r(k) = Bk$
Direct metric distance	$\ r\ ^2 = k^T G k$
Signed scalar address	$\rho(k) = \sigma(k) \sqrt{k^T G k}$
Endpoint direction	$n(k) = Bk / \sqrt{k^T G k}$
Phase cost	$q(n) = n^T G_{\text{phase}} n$
Address-space phase cost	$q(k) = 4(k^T W k) / (k^T G k)$
Path multiplicity	$\Omega_N(k) = \Gamma_N(k) $
Equal-weight path entropy	$S_N(k) = k_B \ln \Omega_N(k)$
Shortest-path multiplicity	$\Omega_{\min} = N_{\min}! / (k_a ! k_b ! k_c !)$
History generating function	$\Omega_N(k) = [z^k] (z_a + z_a^{-1} + z_b + z_b^{-1} + z_c + z_c^{-1})^N$
Entropy matrix	$G_S = B^T (-T) B (-1)$
Large-N entropy	$\ln \Omega_N(R, n) = \text{constant} - [3R^2 / (2N)] n^T G_S n + \dots$
Endpoint weight	$W_N(k) = \Omega_N(k) \exp[-\beta_K q(k)]$
Effective action	$A_N(k) = \beta_K q(k) - \ln \Omega_N(k)$
Effective matrix	$G_{\text{eff}} = \beta_K G_{\text{phase}} + [3R^2 / (2N)] G_S$
Precession	$\dot{n} = -(2/L) n \times G_{\text{phase}} n$
Cost conservation	$dq/dt = 0$
Dwell time	density proportional to $1/ n_{\text{dot}} $
Unified population	P proportional to $\Omega_N \exp[-\beta_K q] C / n \times G_{\text{phase}} n $

Appendix A. Complete 27-node Cell3 ledger

Address	lambda	polarity	kind	x (nm)	y (nm)	z (nm)	rho native
(-1, -1, -1)	-3	-1	corner	-0.176351350144	0.118682429904	0.637311498326	-0.041809908117110
(-1, -1, 0)	-2	+1	edge center	-0.181416008957	-0.076211421278	0.292444294144	-0.021936074880302
(-1, -1, 1)	-1	-1	corner	-0.186480667770	-0.271105272459	-0.052422910037	-0.020735999640663
(-1, 0, -1)	-2	+1	edge center	-0.158685962221	0.222342355125	0.363610955630	-0.028302770281269
(-1, 0, 0)	-1	-1	face center	-0.163750621035	0.027448503943	0.018743751448	-0.010398529391216
(-1, 0, 1)	0	+1	edge center	-0.168815279848	-0.167445347239	-0.326123452733	-0.025117311897496
(-1, 1, -1)	-1	-1	corner	-0.141020574299	0.326002280346	0.089910412934	-0.022802164337825
(-1, 1, 0)	0	+1	edge center	-0.146085233113	0.131108429164	-0.254956791248	-0.020024508730742
(-1, 1, 1)	1	-1	corner	-0.151149891926	-0.063785422018	-0.599823995429	-0.038700006936387
(0, -1, -1)	-2	+1	edge center	-0.012600729109	0.091233925961	0.618567746877	-0.038919790124290
(0, -1, 0)	-1	-1	face center	-0.017665387922	-0.103659925221	0.273700542696	-0.018247102565364
(0, -1, 1)	0	+1	edge center	-0.022730046735	-0.298553776403	-0.071166661485	-0.019152826848624
(0, 0, -1)	-1	-1	face center	0.005064658813	0.194893851182	0.344867204181	-0.024654282949930
(0, 0, 0)	0	+1	body center	0.000000000000	0.000000000000	0.000000000000	0.000000000000000
(0, 0, 1)	1	-1	face center	-0.005064658813	-0.194893851182	-0.344867204181	0.024654282949930
(0, 1, -1)	0	+1	edge center	0.022730046735	0.298553776403	0.071166661485	0.019152826848624
(0, 1, 0)	1	-1	face center	0.017665387922	0.103659925221	-0.273700542696	0.018247102565364
(0, 1, 1)	2	+1	edge center	0.012600729109	-0.091233925961	-0.618567746877	0.038919790124290
(1, -1, -1)	-1	-1	corner	0.151149891926	0.063785422018	0.599823995429	0.038700006936387
(1, -1, 0)	0	+1	edge center	0.146085233113	-0.131108429164	0.254956791248	0.020024508730742
(1, -1, 1)	1	-1	corner	0.141020574299	-0.326002280346	-0.089910412934	0.022802164337825
(1, 0, -1)	0	+1	edge center	0.168815279848	0.167445347239	0.326123452733	0.025117311897496
(1, 0, 0)	1	-1	face center	0.163750621035	-0.027448503943	-0.018743751448	0.010398529391216
(1, 0, 1)	2	+1	edge center	0.158685962221	-0.222342355125	-0.363610955630	0.028302770281269
(1, 1, -1)	1	-1	corner	0.186480667770	0.271105272459	0.052422910037	0.020735999640663
(1, 1, 0)	2	+1	edge center	0.181416008957	0.076211421278	-0.292444294144	0.021936074880302
(1, 1, 1)	3	-1	corner	0.176351350144	-0.118682429904	-0.637311498326	0.041809908117110

Appendix B. Six boundary-plane membership ledger

Channel	Face	Sign	Area (nm ²)	Opposite spacing (nm)	Nine member addresses
a	bc	-1	0.357809539128562	0.329546529709293	(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (-1, 0, -1), (-1, 0, 0), (-1, 0, 1), (-1, 1, -1), (-1, 1, 0), (-1, 1, 1)
a	bc	+1	0.357809539128562	0.329546529709293	(1, -1, -1), (1, -1, 0), (1, -1, 1), (1, 0, -1), (1, 0, 0), (1, 0, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)
b	ac	-1	0.261106235799531	0.451597379724130	(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (0, -1, -1), (0, -1, 0), (0, -1, 1), (1, -1, -1), (1, -1, 0), (1, -1, 1)
b	ac	+1	0.261106235799531	0.451597379724130	(-1, 1, -1), (-1, 1, 0), (-1, 1, 1), (0, 1, -1), (0, 1, 0), (0, 1, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)
c	ab	-1	0.194869165469900	0.605097741514743	(-1, -1, -1), (-1, 0, -1), (-1, 1, -1), (0, -1, -1), (0, 0, -1), (0, 1, -1), (1, -1, -1), (1, 0, -1), (1, 1, -1)
c	ab	+1	0.194869165469900	0.605097741514743	(-1, -1, 1), (-1, 0, 1), (-1, 1, 1), (0, -1, 1), (0, 0, 1), (0, 1, 1), (1, -1, 1), (1, 0, 1), (1, 1, 1)

Appendix C. Exact 10-event minimum-step shell ledger

Minimum steps	Endpoint count	History count	Probability	Mean distance (nm)	Min distance (nm)	Max distance (nm)
0	1	1172556	0.019391932441701	0.000000000000	0.000000000000	0.000000000000
2	18	14770980	0.244285003238836	0.910344607957	0.615518338108	1.584639530085
4	66	24737040	0.409105414570950	1.600727221980	0.855955284386	3.169279060171
6	146	15122430	0.250097343678555	2.297185737704	1.332794175875	4.753918590256
8	258	4203060	0.069510927894630	2.971265986868	1.702907109406	6.338558120341
10	402	460110	0.007609378175329	3.623015653745	2.140966059563	7.923197650427

Appendix D. Phase-cost decile and entropy ledger

Decile	Endpoints	Equal mean q	Equal mean H_fiber	Probability mass	Weighted mean q	Weighted mean H_fiber
1	90	1.807732030660	9.168856694238	0.126241057480	1.817453340855	12.620854167986
2	88	2.310675353217	8.164117643579	0.063335078441	2.259748028725	12.110991443466
3	90	2.926854443606	8.698677519493	0.111754479066	2.891086711316	12.603199696633
4	88	3.544988075327	7.766851497015	0.088773961826	3.654137034775	12.619317426742
5	90	4.154670230333	8.218215443782	0.106519453124	4.156990404452	12.646318191627
6	88	4.829133600660	8.837576688483	0.108703318695	4.822572356108	12.574495876907
7	90	5.682100429926	9.146510258624	0.101554859365	5.827192693245	12.471326584779
8	88	6.546308740629	9.059664669965	0.100539845616	6.514074479677	12.375918072542
9	90	7.440349792917	8.118601074233	0.071087511802	7.339358159362	12.196466992487
10	88	8.301490910307	7.933905948470	0.102098502144	8.324257722923	12.742960809481

Appendix E. Source and integrity ledger

Role	Source file	Epoch	SHA-256
serialized_address_space	K-Field_overdetermined_one-dimensional_serialized_spectral_addresses_space_1785179337(2).json	1785179337	34ccd7ee5c1e7a9071d4cf1ef7164620e067eba133fe33ce43044d0c1556e51b
exact_10_event_propagation	K-Field_10_event_Cell3_propagation_experiment_1785204765.json	1785204765	c446b6f9a9b246ad9e8aabe013a472b92eddbfdae6112210f4bc896cef3363f6
quadratic_phase_cost_and_precession	K-Field_galaxy_gyroscopic_precession_geometry_1784741310.json	1784741310	b78e56ea889d4f790243244d27eacce3d67efd2313a3d09953c22287a4f07249
phase_metric_and_address_coupling_identity	K-Field_single_cell3_lateral_force_reduction_1784985105.json	1784985105	62e4472aacc7df503cc4b5120d7abce97589891e8d22843177ee547e35b71a6f

All geometric figures were regenerated directly from the numerical matrices and address rules printed in this report. All endpoint statistics were computed from the complete exact destination registry. The machine-readable companion contains the full 891-endpoint cost-entropy ledger, the complete mixing sweep, all node and plane records, and hashes of every generated figure.

Archival colophon

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